

Class 5

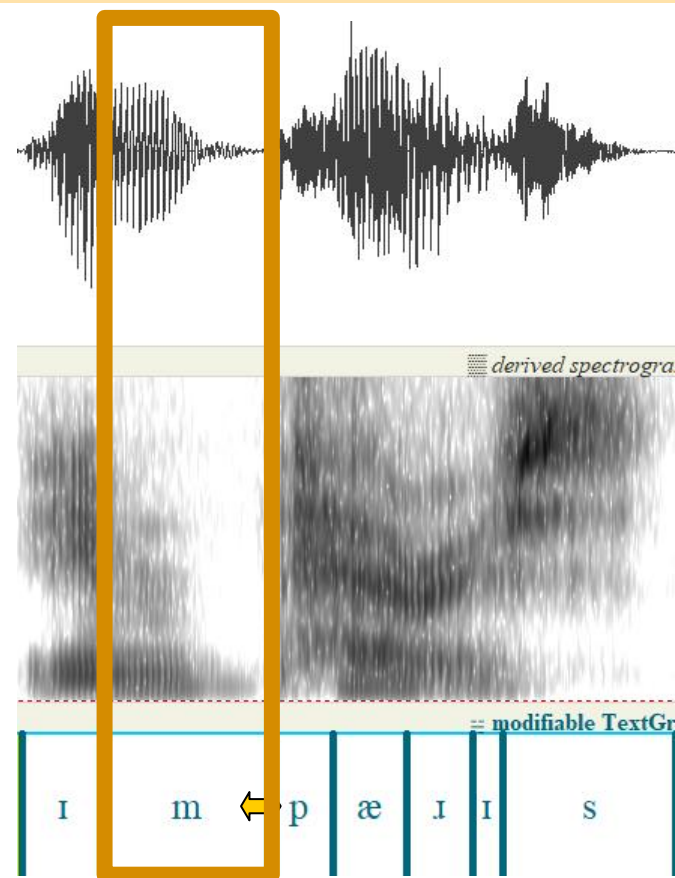
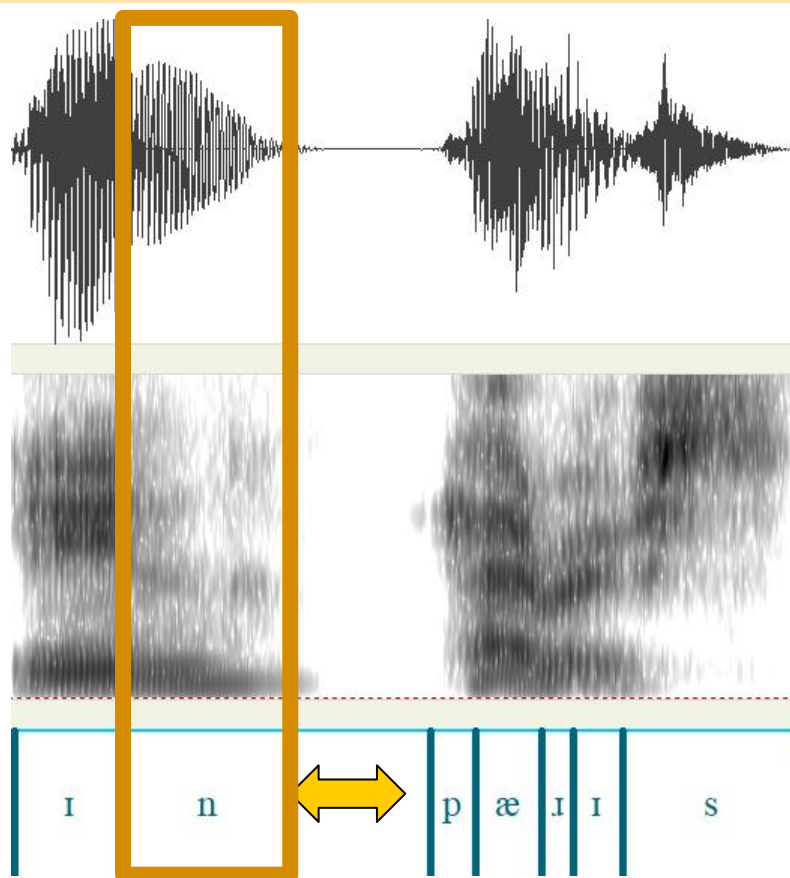
Phonological processes

We will learn the following processes commonly found in many languages:

- ❑ 1. Assimilation
- ❑ 2. Dissimilation
- ❑ 3. Insertion/Epenthesis
- ❑ 4. Deletion/Elision
- ❑ 5. Metathesis

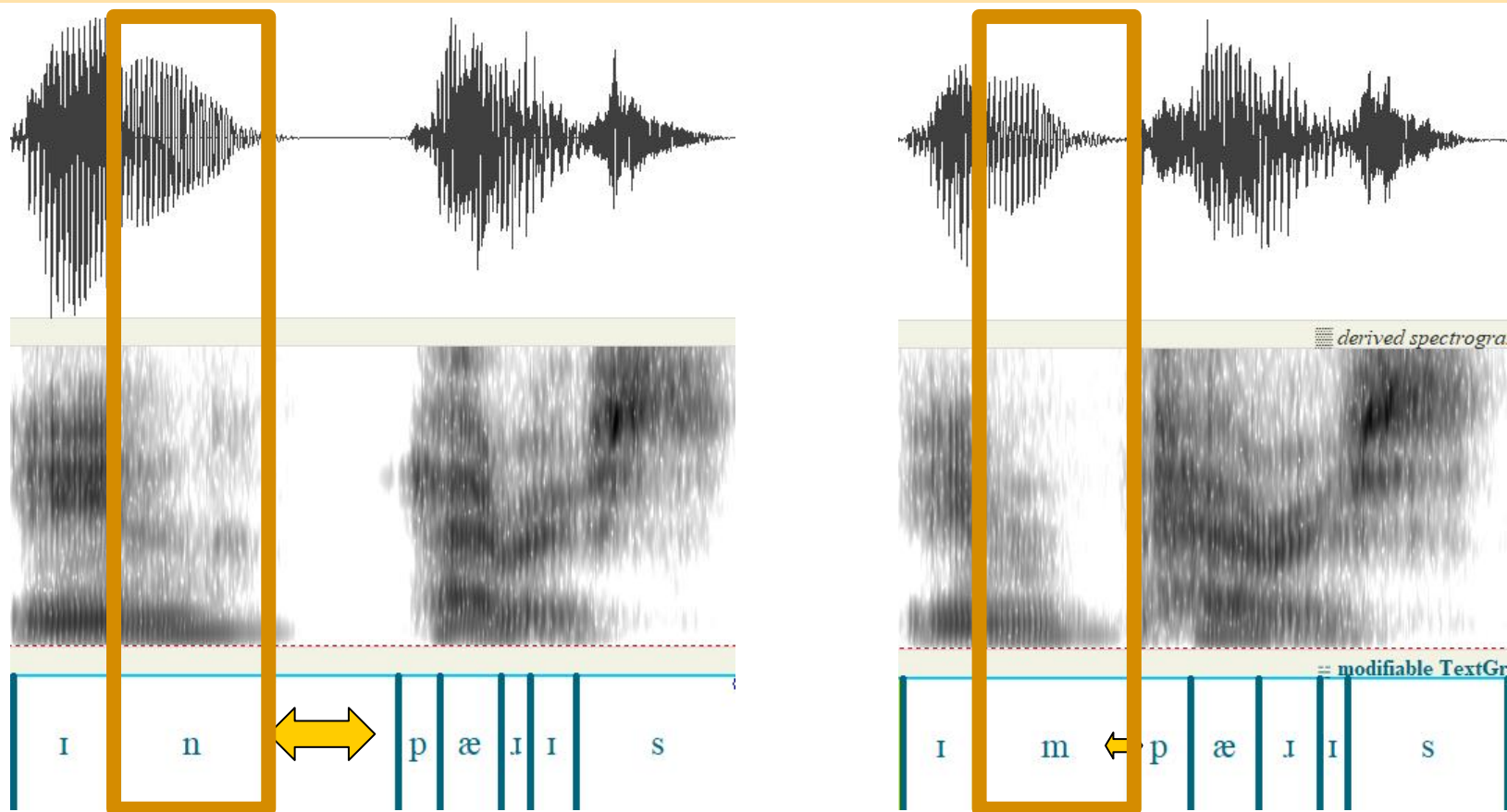
1. Assimilation

▣ **Definition:** One sound becomes more similar to a nearby sound.



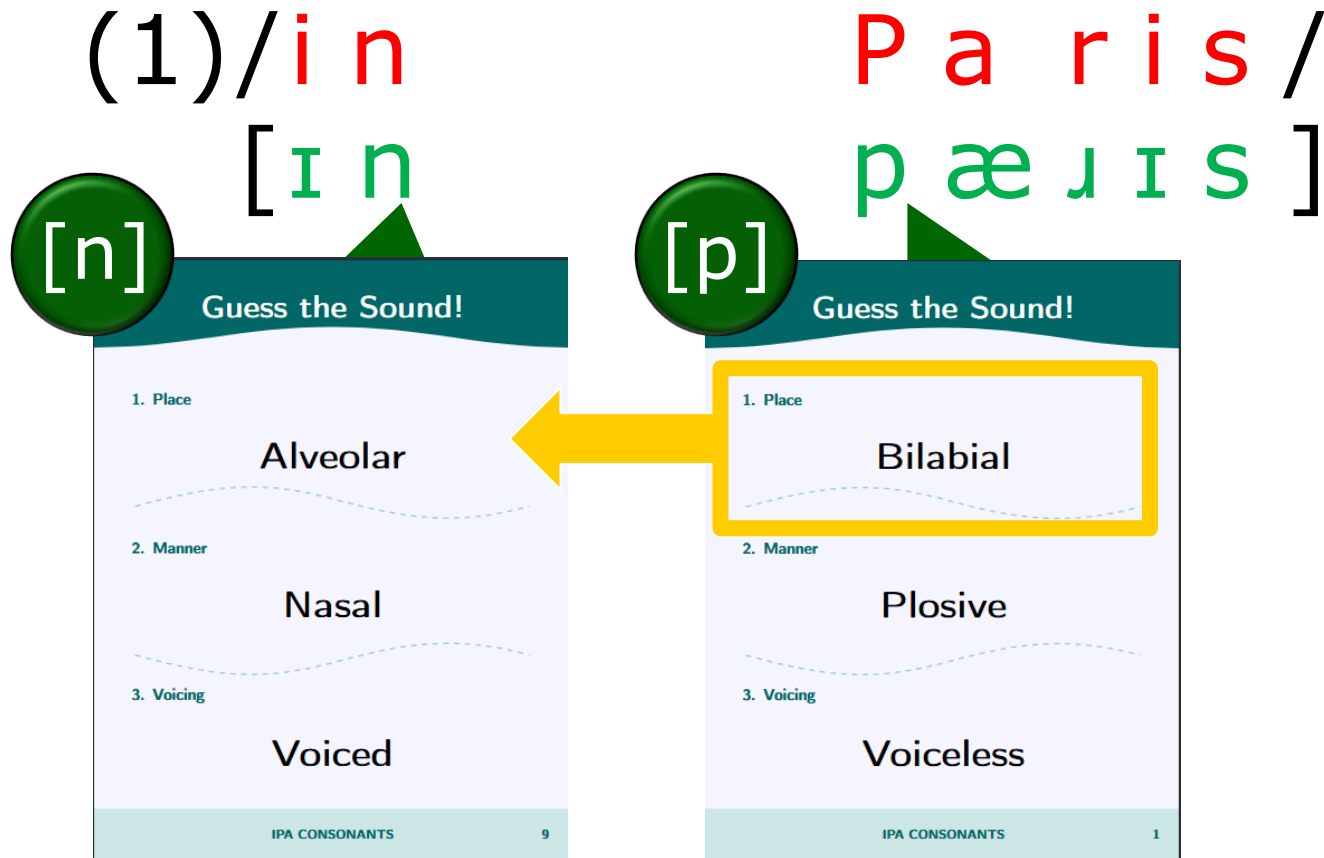
1. Assimilation

- ▣ **Reason:** It reduces articulatory effort by smoothing transitions.



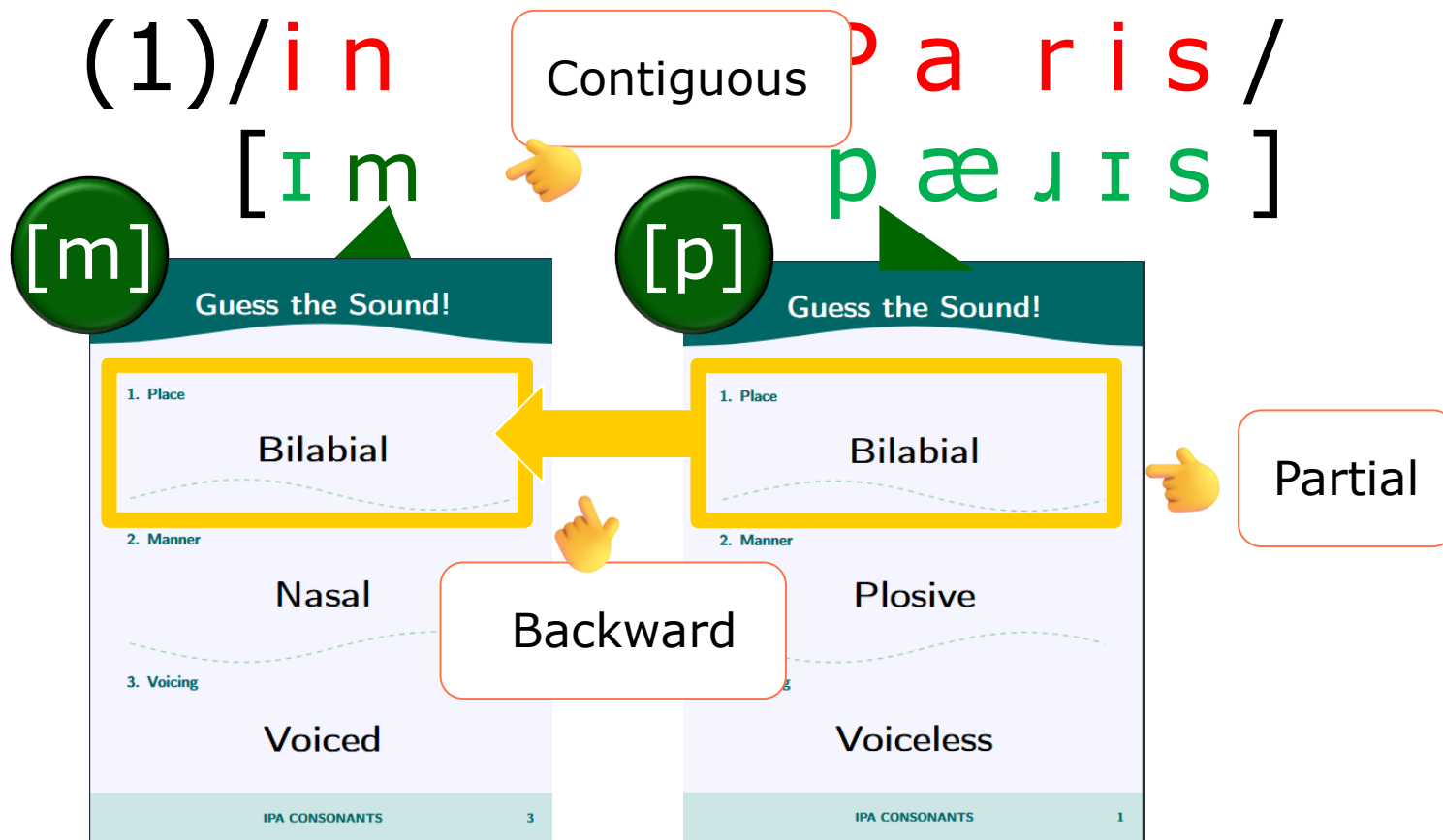
1. Assimilation

- ▣ **Analysis:** a partial, contiguous, backward assimilation



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1. Assimilation

- ▣ **Parameters:** Assimilation is classified by the following three parameters.

Parameter	Type A	Type B
Degree	Partial (sharing some but not all features)	Total (becoming identical)
Proximity	Contiguous (adjacent sounds)	Non-contiguous (sounds at distance)
Direction	Forward (progressive/preservative)	Backward (regressive/anticipatory)

1. Assimilation

- **Hypothesis:** The same should happen with any **other bilabial sounds**.

(1) / i n
[i m

P a r i s /
p æ ɹ i s]

CONSONANTS (PULMONIC)

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	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b		t d			ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ	n			ɳ	ɲ	ŋ	ɴ		
Trill	ʙ		r						ʀ		
Tap or Flap		ⱱ	ɾ			ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative			ɬ ɮ								
Approximant		ʋ	ɹ			ɻ	j	ɰ			
Lateral approximant			l			ɭ	ʎ	ʟ			

1. Assimilation

- ▣ **Hypothesis:** The same should happen with any other bilabial sounds.

(1) /i n/
[ɪ m]

P a r i s /
p æ ɹ ɪ s]

Hypothesis

/n/ → [m] / — $\left[\begin{array}{c} \text{C} \\ [+ \text{bilabial}] \end{array} \right]$ (rapid speech)

Quiz: What word/phrase should we look at?

1. Assimilation

- ▣ **Prediction:** When followed by a bilabial sound it can become an [m].

(1) /i n/
[ɪ m]

P a r i s /
p æ ɹ ɪ s]

(2) /i n/
[ɪ m]

B o s t o n /
b æ s t ə n]

(3) /i n/
[ɪ m]

M a r c h /
m ɑ: ɹ t ʃ]

1. Assimilation

- ▣ **Prediction:** The same should happen with any other bilabial sounds.

(1) /i n/
[ɪ m]

P a r i s /
p æ ɹ ɪ s]

Phonological Rule

/n/ → [m] / — $\left[\begin{array}{c} \text{C} \\ [+ \text{bilabial}] \end{array} \right]$ (rapid speech)

1. Assimilation

▣ **New data:** What happens when it is used with a non-bilabial sound?

(3)Alveolar

T[t]okyo, D[d]enver, S[s]eoul

(4)Post-alveolar

Ch[ʃ]icago, J[dʒ]apan

(5)Velar

C[k]airo, G[g]lasgow

(6)Labiodental

F[f]rankfurt, V[v]enice

in +

1. Assimilation

- ▣ **New data :** When followed by a non-bilabial sound, it should not change.

(3)Alveolar [n]

T[t]okyo, D[d]enver, S[s]eoul

(4)Post-alveolar [ŋ]

Ch[ʃ]icago, J[dʒ]apan

(5)Velar [ŋ]

C[k]airo, G[g]lasgow

(6)Labiodental [m]

F[f]rankfurt, V[v]enice

1. Assimilation

▣ **So far so good?**

Example 1: Germanic umlaut

Backward vowel fronting caused by a subsequent high front sound [i] or [j].

Singular Plural Umlaut
fōt [fo:t] fōt-iz [fo:t-iz] → fēt-iz [fe:t-iz]

[o]

Guess the Vowel!

Guess the Vowel!

[i]

1. Height

Close-mid

2. Frontness

Back

3. Roundedness

Rounded

Tense

1. Height

Close

2. Frontness

Front

3. Roundedness

Unrounded

Tense

Example 1: Germanic umlaut

Backward vowel fronting caused by a subsequent high front sound [i] or [j].

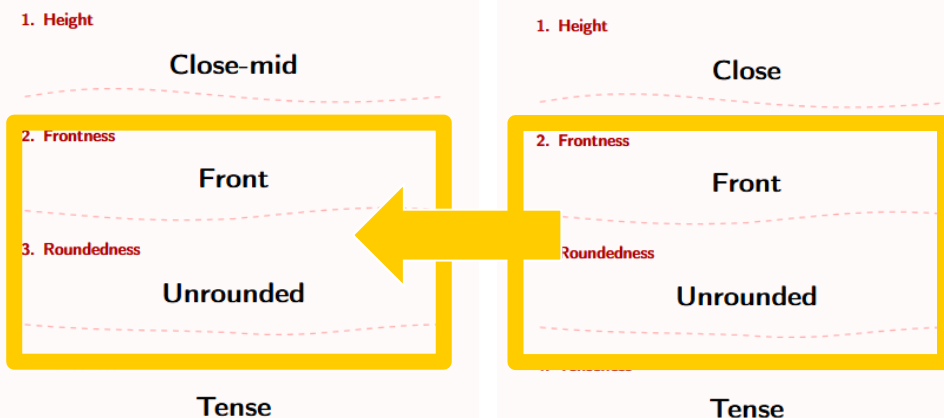
Singular Plural Umlaut
fōt [fo:t] fōt-iz [fo:t-iz] → fēt-iz [fe:t-iz]

[e]

Guess the Vowel!

Guess the Vowel!

[i]



a partial, non-contiguous,
backward assimilation

Example 2: English vs. French

□ obstruction

English

/ə**b**stʌkʃən /

French

/ ɔ**p**stɥɛksjɔ̃ /



2. Dissimilation

▣ **Definition:** One sound becomes more different from a nearby sound.

(1) nature	+ -al	→ natur- al
culture	+ -al	→ cultur- al
education	+ -al	→ edication- al
globe	+ -al	→ glob- al

(2) circle	+ -al	→ circul- ar (*circular)
single	+ -al	→ singul- ar (*singulal)
module	+ -al	→ modul- ar (*modulal)

2. Dissimilation

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- | | | | |
|-----|-----------|-------|---------------------------------|
| (1) | nature | + -al | → natur- al |
| | culture | + -al | → cultur- al |
| | education | + -al | → edication- al |
| | globe | + -al | → glob- al |
| | | | |
| (2) | circle | + -al | → circul- ar (*circular) |
| | single | + -al | → singul- ar (*singulal) |
| | module | + -al | → modul- ar (*modulal) |